



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY

COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

BS Computer Science COURSE INFORMATION SHEET

Session:	Fall-2023
Course Title:	Data Structures and Algorithms
Course Code:	CS-212T
Credit Hours:	3
Semester:	3 rd
Pre-Requisites:	Programming Fundamentals (CS-116T)
Instructor Name:	Mr. Adeel Saeed
Email and Contact Information:	adsaeed@ssuet.edu.pk, HS-12 (Cubical 2)
WhatsApp Group	DS_Fall2023
Office Hours:	Wednesday & Thursday (03:00pm to 05:00pm)
Mode of Teaching:	Synchronous / Asynchronous / Hybrid / Blended

COURSE OBJECTIVE:

This course covers the concepts related to design, analysis and implementation of various data structures and algorithms using C++ programming language. Topics include elementary data structures (arrays, stacks, queues, and linked lists), advanced data structures (including trees and graphs), the algorithms used to manipulate these structures.

COURSE OUTLINE:

Abstract data types, complexity analysis, Big O notation, Stacks (linked lists and array implementations), divide and conquer algorithms, sorting algorithms (selection, insertion, merge, quick, bubble, heap), queue, de-queue, priority queues (linked and array implementations of queues), linked list & its various types, sorted linked list, searching an unsorted array, binary search for sorted arrays, trees and tree traversals, binary search trees, heaps, balanced trees, graphs, breadth-first and depth-first traversal, adjacency matrix and adjacency list implementations, minimum spanning trees.

COURSE LEARNING OUTCOMES (CLOs) and its mapping with Program Learning Outcomes (PLOs):

CLO No.	Course Learning Outcomes (CLOs)	PLOs	Bloom's Taxonomy
1	Understand fundamental data structures as well as algorithms and their complexities.	PLO_1 (Knowledge)	C2 (Understanding)
2	Apply and implementing different data structures in a programming language.	PLO_3 (Design and Development)	C3 (Applying)
3	Solve problem involving graphs, trees and heap.	PLO_2 (Problem Analysis)	C3 (Applying)



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RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tools	CLO-1	CLO-2	CLO-3
Quizzes	3(10%)	3(10%)	4(10%)
Assignments	4(10%)	3(10%)	3(10%)
Midterm Exam	15(50%)	15(50%)	--
Final Exam	10(30%)	10(30%)	30(80%)
Total	31(31%)	31(31%)	38(38%)

GRADING POLICY:

Assessment Tools	Percentage	Marks
Quizzes	10%	10
Assignments	10%	10
Midterm Exam	30%	30
Final Exam	50%	50
TOTAL	100%	100

Recommended Book:

- Weiss, Mark. Data Structures and Algorithm Analysis in (C++,Java), 4th edition, USA, published by Pearson 2014.
- Dale, Nell. Data Structures (C++,Java), 3rd edition, published by Jones and Bartlett.

Reference Books:

- Hemant, Jain. Problem Solving in Data Structures and Algorithm. 2nd edition.
- Drozdek, Adam. Data Structures and Algorithms in C++, 4th edition, USA, published by Cengage Learning 2013.

Benchmark Universities:

- International University: <https://catalogue.surrey.ac.uk/2022-3/module/COM1029>
- National University: <https://www.bahria.edu.pk/wp-content/uploads/2017/06/Data-Structures.pdf>



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LECTURE PLAN

Course Title: Data Structures and Algorithms

Course Code: CS-212T

Week No.	Week Dates	Topics	Required Reading	Key Date
1	09-10-2023 to 13-10-2023	Data Structures and their Operations: Introduction, Overview of Data Structures, Data Structure's operations, Types of Data Structures, common features of using algorithm.	Reading slides material is provided	
2	16-10-2023 to 20-10-2023	Algorithm Analysis: Introduction, Pseudo coding, Control Structures, Analysis of Algorithms, Efficiency of Algorithms, Complexity of Algorithms, Asymptotic Notation/ Big O Notations, Recursion.	Weiss- Chap2 Pg. 29-40 Weiss- Chap1 Pg. 8-12	
3	23-10-2023 to 27-10-2023	Pointers & Arrays: Introduction, Pointer vs. Arrays, Pointer types & operations, 1D and 2D Array, Traversing, insertion, deletion.	Reading slides material is provided	Assignment#1
4	30-10-2023 to 03-11-2023	Sorting Algorithms: Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, Merge Sort.	Nell- Chap10 Pg.588-617	Quiz#1
5	06-11-2023 to 10-11-2023	Linear and Binary Search Linked Lists: Introduction, Types of Linked List, Insertion and Deletion in singly LL.	Nell- Chap10 Pg.619-621 Weiss- Chap3 Pg. 77-91	
6	13-11-2023 to 17-11-2023	Doubly and Circular Linked Lists: Doubly Linked List, Circular Linked List.	Nell- Chap6 Pg.333-344	
7	20-11-2023 to 24-11-2023	Stacks: Introduction, Stack Model, Array & Linked List representation of Stacks. Polish Notation Prefix, Infix, Postfix.	Weiss-Chap3 Pg. 103-110	Assignment#2
8	27-11-2023 to 01-12-2023	Queues: Introduction, Queue Model, Array & Linked List implementation of Queues.	Weiss-Chap3 Pg. 112-115	Quiz#2
Midterm Examination (04-12-2023 to 09-12-2023)				



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10	11-12-2023 to 15-12-2023	Trees: Basic Terminologies, Binary Trees, Array/Linked List representation of binary trees in memory, Traversing Binary Trees, Pre, In, and Post-order traversal	Weiss-Chap4 Pg. 121-125	
11	18-12-2023 to 22-12-2023	Binary Search Trees: Binary Search Trees, Searching, Inserting, and Deleting in Binary Search Trees, Efficiency of BST	Weiss-Chap4 Pg. 132-141	
12	25-12-2023 to 29-12-2023	AVL Tree, Creating AVL Trees, Balancing AVL Tree by single or double rotations. Minimum Spanning Trees: Shortest path algorithms (Prim's Algorithm, Dijkstra's Algorithm)	Weiss-Chap4 Pg. 144-149 Weiss-Chap9 Pg. 391 Pg.413-417	Assignment#3
13	01-01-2024 to 05-01-2024	Graphs: Basic Terminologies, Graph Theory, Directed, Undirected Graphs, Sequential representation of graphs, Adjacency Matrix. Link List representation of graphs, Operations on Graphs, Graph Traversals (Depth-First Traversal, Breadth First Traversal).	Nell-Chap9 Pg.546-570	
14	08-01-2024 to 12-01-2024	Hashing Types of hashing: division, mid square, folding. Hash collision Linear Probing	Weiss-Chap5 Pg. 193-208	Quiz#3
15	15-01-2024 to 19-01-2024	Heap: Binary Heaps, Heap creation and operations, Min and Max Heap, Heap sort, Insertion and deletion in heap.	Weiss-Chap6 Pg. 245-262	
16	22-01-2024 to 26-01-2024	Data Compression: Huffman's Algorithm. Introduction to Knapsack Problem: Branch and Bound, Greedy Algorithm.	Weiss-Chap10 Pg. 449-453	
Final Examination (30-01-2024 to 10-02-2024)				

Teacher's Name and Signature: Adeel Saeed Date: 25-09-2023Chairperson's Name and Signature: Dr. Waleej HaiderDate: 25-09-2023